
SmartJourney

Smart Tourism with Multi agent collaboration Software Requirements Specification

Version 1.0

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Group No 08

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Index No	Contribution
230247T	Design Constraints, On-line User Documentation and Help System Requirements, Purchased Components, Interfaces (User, Hardware, Software, Communication)
230279R	Functionality, Usability, Reliability, Performance & security, Supportability, Applicable standards
230281P	Introduction, overall description, Supporting Information, Database Requirements, Licensing, Legal, Copyright, and Other Notices

SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

Revision History

Date	Version	Description	Author
09/Aug/26	1.0	Initial release of the Software Requirements Specification for SmartJourney.	SmartJourney Team

SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

Table of Contents

Revision History 2

Table of Contents 3

1. Introduction 6

 1.1 Purpose..... 6

 1.2 Scope..... 6

 1.3 Definitions, Acronyms, and Abbreviations 7

 1.4 References..... 8

 1.5 Overview..... 8

2. Overall Description 9

3. Specific Requirements 10

 3.1 Functionality 10

 3.1.1 User Registration 10

 3.1.2 User Login 11

 3.1.3 Reset Password 11

 3.1.4 Change Password..... 11

 3.1.5 Manage User Account..... 11

 3.1.6 Generate AI Travel Itinerary..... 12

 3.1.7 Explore Destinations..... 12

 3.1.8 Budget Tracker..... 12

 3.1.9 Subscription Management 12

 3.1.10 Manage Notifications..... 13

 3.1.11 Manage Attractions, Restaurants, Events and Travel Options 13

 3.1.12 Verify Travel Listing 13

 3.1.13 Manage Users..... 13

 3.1.14 View System Analytics..... 13

 3.2 Usability 13

 3.2.1 AI Conversations 13

 3.2.2 Interactive Navigation..... 14

 3.2.3 Personalization..... 14

 3.2.4 Error Prevention and User Feedback 14

 3.2.5 Accessibility..... 14

 3.2.6 Responsiveness 14

SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

3.3 Reliability.....	14
3.3.1 Availability	14
3.3.2 Mean Time Between Failure (MTBF)	15
3.3.3 Mean Time To Repair (MTTR)	15
3.3.4 Accuracy	15
3.3.5 Maximum Bugs or Defect Rate	15
3.3.6 Bugs or Defect Rate	15
3.4 Performance and Security	16
3.4.1 Response Time	16
3.4.2 Throughput.....	16
3.4.3 Capacity	16
3.4.4 System Up Time	16
3.4.5 Resource Utilization.....	17
3.4.6 User Data Protection	17
3.5 Supportability.....	17
3.5.1 Maintainability	17
3.5.2 Scalability	17
3.5.3 Configurability	17
3.5.4 Documentation	17
3.5.5 Software Updates	17
3.5.6 Installability	18
3.6 Design Constraints	18
3.6.1 Standards Compliance	18
3.6.2 Hardware Limitations	18
3.6.3 Architectural and Design Constraints	18
3.7 On-line User Documentation and Help System Requirements.....	18
3.8 Purchased Components	19
3.9 Interfaces	20
3.9.1 User Interfaces	20
3.9.2 Hardware Interfaces	31
3.9.3 Software Interfaces	31
3.9.4 Communications Interfaces	31
3.10 Database Requirements.....	32
3.11 Licensing, Legal, Copyright, and Other Notices	33

SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

3.12 Applicable Standards 34

4. Supporting Information..... 34

5. References..... 34

SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

Software Requirements Specification

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define, in detail, the functional and non-functional requirements of SmartJourney, an AI-powered smart tourism assistant platform delivered as a combined web and mobile application. This document specifies the external behavior of the system, including the features available to travelers, administrators, and content providers, as well as the constraints, performance expectations, and design decisions that govern its implementation.

This SRS is intended to serve as a shared reference point for the development team and project stakeholders, ensuring a consistent understanding of what the system is expected to do before implementation begins. It captures the requirements for both the web and mobile applications, as well as the Agentic AI System that supports itinerary generation and travel recommendations. Subsequent design, development, and testing of the system are expected to be carried out against the requirements defined in this document.

1.2 Scope

This SRS applies to SmartJourney, an AI-powered smart tourism assistant platform consisting of a web application, a mobile application (Android and iOS), and a supporting Agentic AI System. The platform is intended to centralize the trip-planning process for travelers by combining a conversational AI interface with a curated, admin-verified database of hotels, restaurants, attractions, and local events.

The scope of this document covers the requirements for three primary user roles: registered travelers, who use the platform to plan trips, generate AI-assisted itineraries, track travel budgets, and manage their accounts; administrators, who manage and verify platform content and monitor system usage through an administration dashboard; and the Agentic AI System itself, which interprets natural-language travel requests and autonomously generates, refines, and optimizes personalized itineraries based on verified internal data.

This specification also covers the functional and non-functional requirements of the platform, including integrations with external services such as AI providers, mapping and weather services, cloud infrastructure, and other supporting technologies required for system operation.

SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

1.3 Definitions, Acronyms, and Abbreviations

Definitions

Term	Definition
Agentic AI System	The AI-based subsystem responsible for interpreting natural-language travel requests and autonomously generating, refining, and optimizing personalized itineraries.
NestJS	A backend application framework used to build the platform's REST API services.
Next.js	A web application framework used to build the platform's frontend.
Flutter	A cross-platform framework used to build the platform's mobile application.
PostgreSQL	A relational database management system used to store the platform's data.
FastAPI	A Python web framework used to implement the platform's AI service layer.
LangChain	A software framework used to orchestrate interactions between the Agentic AI System and the Gemini API.
bcrypt	A password hashing algorithm used to securely store user passwords in the database.

Acronyms and Abbreviations

Acronym	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
AWS	Amazon Web Services
COTS	Commercial Off-The-Shelf
FCM	Firebase Cloud Messaging
GPS	Global Positioning System
GUI	Graphical User Interface
HTTPS	Hypertext Transfer Protocol Secure
IEEE	Institute of Electrical and Electronics Engineers

SmartJourney		Version: 1.0
Software Requirements Specification		Date: 09/Aug/26
Acronym	Meaning	
ISO/IEC	International Organization for Standardization / International Electrotechnical Commission	
JWT	JSON Web Token	
KLOC	Thousand Lines of Code	
LKR	Sri Lankan Rupee	
MTBF	Mean Time Between Failure	
MTTR	Mean Time To Repair	
NIC	Network Interface Card	
PDPA	Personal Data Protection Act	
RDS	(Amazon) Relational Database Service	
REST / RESTful	Representational State Transfer	
SES	(Amazon) Simple Email Service	
SRS	Software Requirements Specification	
TCP/IP	Transmission Control Protocol / Internet Protocol	

1.4 References

- [1] Institute of Electrical and Electronics Engineers, "IEEE Std 1063-1987, IEEE Standard for Software User Documentation," IEEE, 1987.
- [2] International Organization for Standardization / International Electrotechnical Commission / Institute of Electrical and Electronics Engineers, "ISO/IEC/IEEE 29148:2018, Systems and software engineering – Life cycle processes – Requirements engineering," ISO/IEC/IEEE, 2018.

1.5 Overview

The remainder of this document is organized as follows:

Section 2, Overall Description, provides background on the system as a whole, including its product perspective, core functions, intended users, constraints, and assumptions, without stating specific requirements.

Section 3, Specific Requirements, contains the detailed functional and non-functional requirements of the system, organized by feature area. This includes functionality (3.1), usability (3.2), reliability (3.3), performance and security (3.4), supportability (3.5), design constraints (3.6), on-line documentation and help system requirements (3.7), purchased components (3.8), interfaces

SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

(3.9), database requirements (3.10), licensing and legal notices (3.11), and applicable standards (3.12).

Section 4, Supporting Information, describes how this document is organized and notes that interface mockups referenced throughout Section 3 are provided inline within their respective subsections rather than as a separate appendix.

2. Overall Description

Product Perspective

SmartJourney is a new, self-contained software product rather than a component of, or replacement for, an existing system. It follows a client-server architecture, in which the web and mobile client applications communicate with a NestJS backend[5] through RESTful APIs. The backend interfaces with a PostgreSQL database[8] (hosted via Amazon RDS[10]) and with a separately developed Agentic AI System, built using Python, FastAPI, and LangChain[9], which orchestrates itinerary generation through the Gemini API[1]. The platform also depends on several external services for core functionality, including the Google Maps Platform for navigation and mapping[2], the OpenWeather API for weather data[3], and Firebase Cloud Messaging for push notifications[4], as detailed in the Interfaces and Purchased Components sections.

Product Functions

At a high level, SmartJourney provides the following categories of functionality, detailed fully in Section 3.1:

- Account creation, authentication, and profile management for travelers.
- AI-assisted, conversational itinerary generation based on user-stated destinations, budgets, and preferences.
- Explore destinations and filtering of curated attractions, restaurants, and events.
- Travel budget tracking and expense management across saved trips.
- Subscription plan management for premium features.
- Configurable notifications for trip reminders, weather alerts, and budget alerts.
- Administrative tools for managing, verifying, and publishing travel-related content, as well as monitoring platform usage and analytics.

User Characteristics

The system is designed for two primary classes of users. Registered travelers are expected to represent the general public, with no assumed technical expertise; the conversational, chat-based interface is intended to accommodate users regardless of prior experience with travel-planning tools. Administrators are expected to be operationally familiar with the platform's content

SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

categories and verification workflow, interacting with the system primarily through the administration dashboard described in Section 3.9.1.

Constraints

The system's design and implementation are constrained by the prescribed technology stack (Next.js, Flutter, NestJS, and PostgreSQL, as specified in Section 3.6), and by the requirement to comply with the Sri Lanka Personal Data Protection Act, No. 9 of 2022 (PDPA)[13]. The mobile application is limited to the platforms specified in Section 3.6. Since core functionality depends on external APIs and cloud-hosted infrastructure, client devices require a stable internet connection to use the platform, as noted in Section 3.9.2.

Assumptions and Dependencies

The requirements defined in this document assume continued availability and accessibility of all third-party services the platform depends on, including the Gemini API[1], Google Maps Platform[2], OpenWeather API[3], and AWS[10] infrastructure, within their expected usage tiers and pricing models as outlined in the Purchased Components section. It is also assumed that administrators will review and verify submitted content in a timely manner, since the reliability of recommendations presented to travelers depends directly on this verification process, as discussed in Section 3.1.12.

Requirements Subsets

All requirements defined in Section 3 apply to the current release of the system as a single, unified set; this document does not currently define separate requirement subsets for phased or incremental releases.

3. Specific Requirements

3.1 Functionality

3.1.1 User Registration

A user needs to register for the system to access all the services of the web application according to the role of the user (admin/traveler). Unregistered users are restricted from storing payment methods, personalized recommendations, and viewing booking history.

When registering to the system the user must give their email address and a strong password (that must contain at least one upper case letter, one lower case letter, one digit, one special character, and must be 8-12 characters long) after agreeing to terms and policies; the system will send a verification code to the email address for login to the system, or the user can sign up using a Google account.

SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

After the account is successfully created, the system allows the user to complete their travel profile by providing optional personalization details including name, profile picture, preferred language and currency, country/region, travel interests (adventure, food, culture), preferred travel style (solo, couple, family, friends), travel likes and dislikes, and other preferred specifications when travelling.

3.1.2 User Login

Registered users can log into the system using their registered email address and password or through a Google account. The system validates the entered credentials. If authentication is successful, the user is redirected to the personalized dashboard with the facility to change or update the travel profile.

If incorrect credentials are entered, the system displays an appropriate error message and allows the user to try again.

Users may also choose the “Remember Me” option so that they remain signed in on trusted devices.

3.1.3 Reset Password

When a user forgets their password, they can click the ‘Forgot password?’ link, after which the system will send a link to the given email to reset the password to a new one, subject to the password restrictions described above. After that, the user is able to log in to the system with the email and new password.

3.1.4 Change Password

Users can change their password at any time from the Profile page after logging into the system. The user must enter the current password before changing it; the system then checks whether the new password satisfies the password strength requirements. If all requirements are satisfied, the password is updated securely in the database and the user receives a confirmation message.

3.1.5 Manage User Account

Registered users can manage their account through the Account Settings page, accessible from the settings menu after logging into the system.

Users can update their username and phone number, and can also enable or disable location access according to their privacy preferences.

The account page also allows users to change their passwords. After updating the required information, the user can save the changes; the system validates the entered information before updating the user's account.

SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

3.1.6 Generate AI Travel Itinerary

Registered users can generate personalized travel itineraries using the Main Dashboard page.

Users first provide their travel destination, travel duration, budget, travel companions, and any special requirements using natural language.

The AI analyzes the request and generates a complete day-by-day itinerary containing recommended attractions, restaurants, hotels, and activities. The generated itinerary is displayed together with an interactive route map.

Users are able to continue the conversation to modify the itinerary, request weather-aware recommendations, change the budget, or replace activities. Once satisfied, users can save the itinerary for future use.

3.1.7 Explore Destinations

Any user can explore destinations through the Explore page.

Users can search destinations using the search bar or browse recommended hotels, tourist attractions, hidden gems, and local cultural events.

Users can apply filters to narrow search results according to the destination category.

3.1.8 Budget Tracker

Registered users can manage their travel expenses through the Budget Tracker page.

Users can define a total budget for each trip and record expenses under different categories including accommodation, transportation, food, and activities.

The system automatically calculates the total expenses, remaining budget, daily average expenditure, and category-wise spending.

Users receive visual indicators showing whether the trip is within budget or exceeding the planned budget.

3.1.9 Subscription Management

Registered users can manage their subscription plans through the Subscription page.

Users can view available subscription plans together with their features and monthly charges.

The system allows users to upgrade, downgrade, or maintain their current subscription plan.

After a successful subscription update, the user's account automatically gains access to the corresponding premium features.

SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

3.1.10 Manage Notifications

Registered users can configure notification preferences through the Notification Settings page.

Users can enable or disable trip reminders, weather alerts, budget alerts, push notifications, and email notifications. Users may also configure notification sounds and notification volumes. The system saves all notification preferences and delivers future notifications according to the selected settings.

3.1.11 Manage Attractions, Restaurants, Events and Travel Options

Administrator users can manage travel-related content through the administration dashboard.

Administrators can add, update, or remove attractions, restaurants, events, and travel options available in the system. Each content item contains information including destination, category, description, images, and status.

3.1.12 Verify Travel Listing

Administrator users can verify travel listings submitted by content providers.

The system maintains a verification queue containing newly submitted attractions, restaurants, events, and travel services. Administrators may approve or reject each submission after reviewing its details. Only approved listings are published for users.

3.1.13 Manage Users

Administrator users can manage registered users and content providers through the administration dashboard. Administrators can view user information (excluding password), search user accounts, monitor account activities, and manage user access privileges.

3.1.14 View System Analytics

Administrator users can monitor overall system performance through the analytics dashboard.

The dashboard displays statistics including the number of registered travelers, generated itineraries, pending listing verifications, and subscription revenue. The system also presents charts showing AI itinerary generation trends and other platform usage statistics to assist administrators in monitoring system performance.

3.2 Usability

3.2.1 AI Conversations

The AI travel assistant provides a natural and interactive way for users to plan their trips. Users can communicate with the chatbot using everyday language instead of filling out lengthy forms. During the conversation, the AI asks relevant questions to understand user preferences such as

SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

destinations, travel dates, budget, travel style, and accommodation preferences before generating a personalized itinerary. Users can continue the conversation to modify or regenerate any part of the itinerary without restarting the planning process.

3.2.2 Interactive Navigation

The system provides a simple and consistent navigation structure throughout the application. Major features including AI Trip Planner, Explore, Saved Itineraries, Budget Tracker, Notifications, and Account Settings can be accessed directly through the navigation menu. The interactive map allows users to visualize destinations and travel routes while selecting locations to obtain additional travel information.

3.2.3 Personalization

SmartJourney provides a personalized travel planning experience based on individual user preferences. The system remembers user interests, preferred destinations, budgets, and previous itineraries to generate more relevant travel recommendations. Saved itineraries and user preferences allow travelers to continue planning future trips.

3.2.4 Error Prevention and User Feedback

The system validates user inputs before processing travel requests. Invalid travel dates, incomplete information, or incorrect email addresses generate error messages that explain the problem.

3.2.5 Accessibility

The interface uses readable fonts, clear icons, and color contrast to improve readability for different users.

3.2.6 Responsiveness

The application is operated across desktop computers, tablets, and mobile devices. The layout automatically adjusts according to the screen size while maintaining readability and accessibility.

3.3 Reliability

3.3.1 Availability

The SmartJourney platform should be available at least 99% of the time. Users should be able to access system features such as AI itinerary generation, destination exploration, saved itineraries, budget tracking, and account management whenever required.

Any maintenance should be scheduled during minimal user activity and should be announced in advance.

All data should be backed up to enable restoration in the event of a hardware failure, database corruption, or hacking attack.

SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

3.3.2 Mean Time Between Failure (MTBF)

The system is expected to run 24/7. However, 4 downtimes per month are acceptable.

Total working time per year = 24×365 hr

Total downtime per year = $4 \times 12 = 48$ hr

MTBF = $(24 \times 365) / 48 = 182.5$ hr

Proper monitoring, regular maintenance, and testing should be carried out to minimize unexpected failures and improve system stability.

3.3.3 Mean Time To Repair (MTTR)

When a system failure happens, the application should be restored in less than 6 hours. Any maintenance should be prescheduled. The system should inform users about the maintenance, and system unavailability time should be less than 12 hours.

3.3.4 Accuracy

The system should generate travel itineraries and recommendations based on the information provided by users together with verified tourism data stored in the system. Budget calculations, travel schedules, and destination information should be processed accurately to provide reliable travel plans. Weather forecasts and maps should be updated regularly to maintain accuracy.

3.3.5 Maximum Bugs or Defect Rate

The maximum acceptable software defect rate for SmartJourney should not exceed 1 defect per thousand lines of code (1 defect/KLOC).

3.3.6 Bugs or Defect Rate

Minor Bugs

Most of the time, these bugs will not result in any noticeable breakdown of the system. Examples include:

- Travel destination images (hotels and viewpoints) fail to load.
- Misalignments in icons and buttons.
- Spelling or grammatical errors.

Significant Bugs

These bugs will affect some functions of the system. Most of the time, these bugs will halt some functionalities or make them inefficient. Examples include:

- Notification emails or trip reminders are delayed or not delivered.

SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

- Some verified attractions are not displayed on the Explore page.
- Weather information is outdated.
- Budget calculations display incorrect estimated travel expenses.
- The map does not display some recommended destinations or routes correctly.

Critical Bugs

Critical bugs prevent users from accessing the system or using its functionalities. Examples include:

- Database becomes unavailable.
- Server crashes.
- Saved data is lost or cannot be accessed from the database.
- Relevant subscription upgrade is not available after completing payment.
- Users are unable to log in or register.

3.4 Performance and Security

3.4.1 Response Time

The system should respond quickly to user requests to provide a smooth travel planning experience. User authentication, destination searches, and page navigation should normally be completed within 2 seconds. During processing, the system should display loading indicators to inform users that the request is being processed.

3.4.2 Throughput

The system should support multiple users performing activities simultaneously. The system should be capable of processing at least 50 user requests per second without significant degradation in performance.

3.4.3 Capacity

The web app will be hosted on the Amazon Web Services (AWS) free tier, which can handle 20,000 concurrent users. This limitation is only valid for the web app; there are no limitations for Android app installation. However, there is a limitation on the requests made by both the web and mobile apps: the maximum request rate for the REST API from both the Android and web apps should not exceed 1000 requests.

3.4.4 System Up Time

The system should have at least 99% uptime. Users should be informed about maintenance at least one week in advance. The maintenance period should be less than 12 hours.

SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

3.4.5 Resource Utilization

The system should utilize server resources efficiently to maintain stable performance. Database queries, AI service requests, and map rendering should be optimized to minimize unnecessary memory usage, processor utilization, and network communication.

3.4.6 User Data Protection

All passwords supplied by users should be hashed before being stored in the database. Password field types should be used to retrieve passwords from users, and passwords should not be displayed on the GUI. User data should be encrypted before being sent over the network.

3.5 Supportability

3.5.1 Maintainability

SmartJourney follows a Client-Server Architecture, where the frontend communicates with the backend through REST APIs. The backend is organized into separate layers to simplify maintenance, testing, and future enhancements. This architecture allows developers to modify individual system components without affecting the overall functionality of the platform.

3.5.2 Scalability

The database is designed to support future expansion by allowing new tourism destinations, accommodation providers, restaurants, and other travel-related information to be added without affecting existing user data or system operations. AI itinerary generation, tourism information management, user authentication, and notification services are implemented as independent components, which also supports system scalability.

3.5.3 Configurability

The framework should be executed using a platform-independent stack, so that changing the environment of the system does not change or crash system features.

3.5.4 Documentation

Technical documentation, API documentation, and user guides should be maintained throughout the development lifecycle. Documentation should be updated whenever new features or system modifications are introduced to support future maintenance and development activities.

3.5.5 Software Updates

The system should support regular software updates to introduce new features and improve AI performance. Any saved data or tourism information should remain unaffected during and after the update process.

SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

3.5.6 Installability

This system has both web and mobile applications. Web applications do not require any installation. On mobile platforms, the app can be used on all Android devices, with expansion to the Apple App Store planned for the future.

3.6 Design Constraints

3.6.1 Standards Compliance

- **Data Privacy and Protection:** The system must comply with the Sri Lanka Personal Data Protection Act, No. 9 of 2022 (PDPA)[13]. The platform must collect only the personal information necessary for system functionality and ensure user data is not disclosed to third parties without appropriate authorization.
- **Security Standards:** Session management and role-based access control must be implemented using JSON Web Tokens (JWT)[12]. Additionally, user passwords must be encrypted using the bcrypt hashing algorithm before being stored in the database.
- **Development Standards:** Backend communications and data synchronization between the web and mobile applications must be handled via RESTful API services.

3.6.2 Hardware Limitations

- **Client-Side Connectivity:** Because the system's core features depend on external APIs and cloud-hosted services, client devices require internet connectivity to communicate with the AWS infrastructure and the Agentic AI System.
- **Mobile Platform Constraints:** The mobile application is constrained to operating on Android or iOS platforms.

3.6.3 Architectural and Design Constraints

- **Prescribed Technology Stack:** The system architecture must utilize Next.js[6] for the frontend web application, Flutter[7] for the mobile application, NestJS[5] for the backend, and PostgreSQL[8] for the database.
- **AI Service Decoupling:** The Agentic AI System must be developed independently from the main backend using Python, FastAPI, and LangChain to orchestrate calls to the Gemini API.

3.7 On-line User Documentation and Help System Requirements

- **Conversational Interface:** The primary method for user interaction is a chat-based interface designed to interpret natural language requests to generate personalized travel itineraries.

SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

- **Account Recovery:** The system must support user account recovery, utilizing Amazon Simple Email Service (Amazon SES)[11] to send email verification links for password resets.
- **System Notifications:** The system must be capable of delivering travel-related notifications to the mobile application via Firebase Cloud Messaging[4].

3.8 Purchased Components

- **Cloud Infrastructure:** The system requires Amazon Web Services (AWS)[10] for cloud deployment. This includes Amazon Relational Database Service (RDS)[10] for hosting the PostgreSQL database, Amazon SES[11] for system emails, and Amazon CloudWatch[10] for application logs and performance monitoring.
- **External APIs:** The platform integrates usage-based external services, specifically the Gemini API[1] for natural language processing, the Google Maps Platform[2] for interactive maps, and the OpenWeather API[3] for dynamic weather forecasting.
- **Deployment and Licensing:** Releasing the platform requires purchasing a custom .com domain name (estimated at LKR 5,000/year) and paying a \$25 registration fee for a Google Play Console account to deploy the Android application.

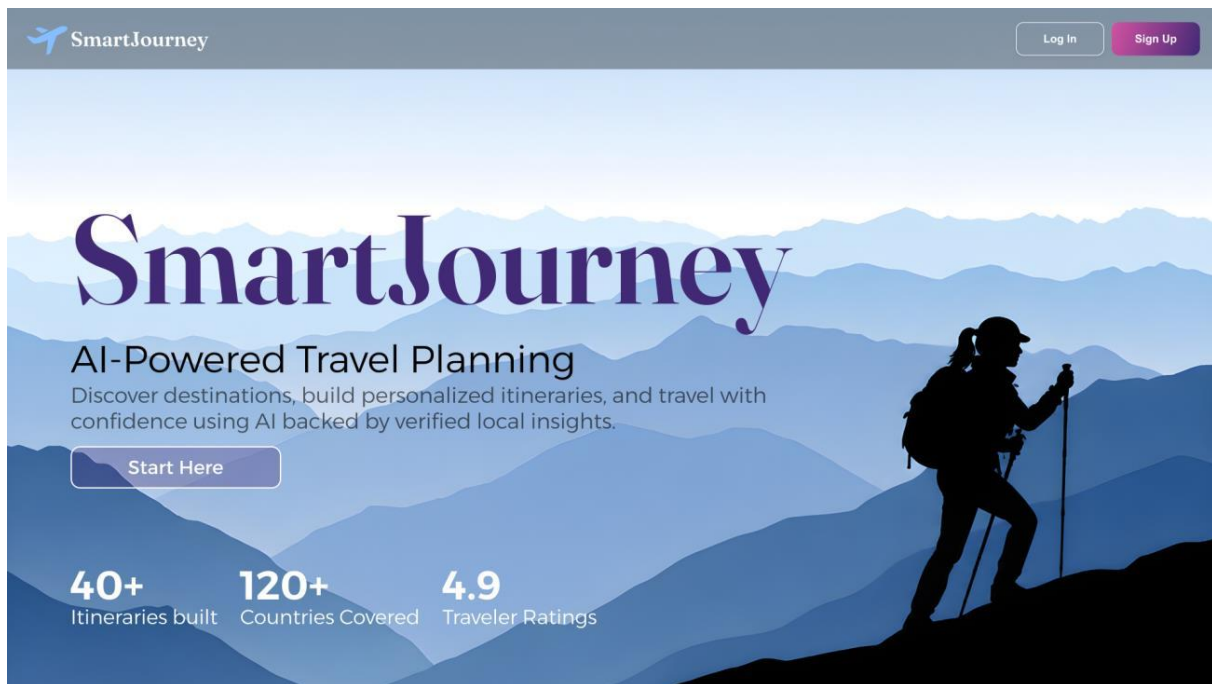
SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

3.9 Interfaces

3.9.1 User Interfaces

3.9.1.1 Landing Page

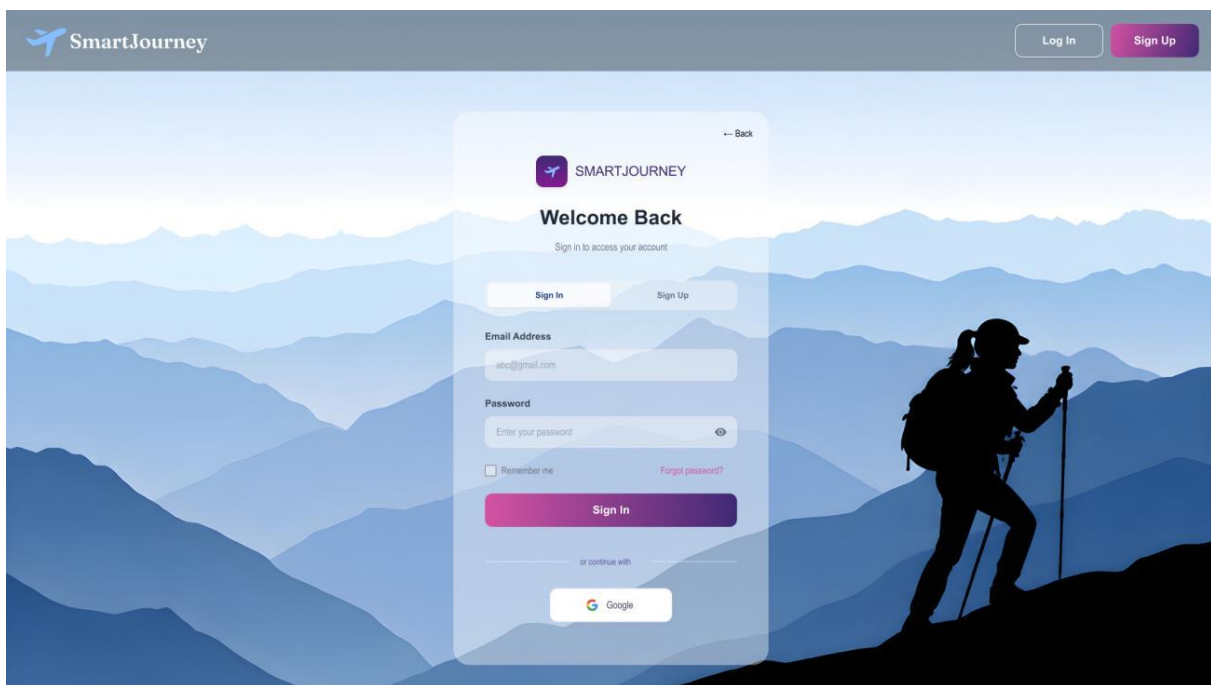
This is the initial page that every user can see. On the top navigation bar, the logo and website name “SmartJourney” are displayed along with the Log In and Sign Up buttons on the left corner. In the next section, users can start planning their trip by clicking the Start Here button, which leads to the Log In page. The middle part includes the main heading, a brief description, and an illustrative background image of a hiker. Below, there are several statistics displaying valuable information such as the number of itineraries built, countries covered, and traveler ratings.



SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

3.9.1.2 Login Page

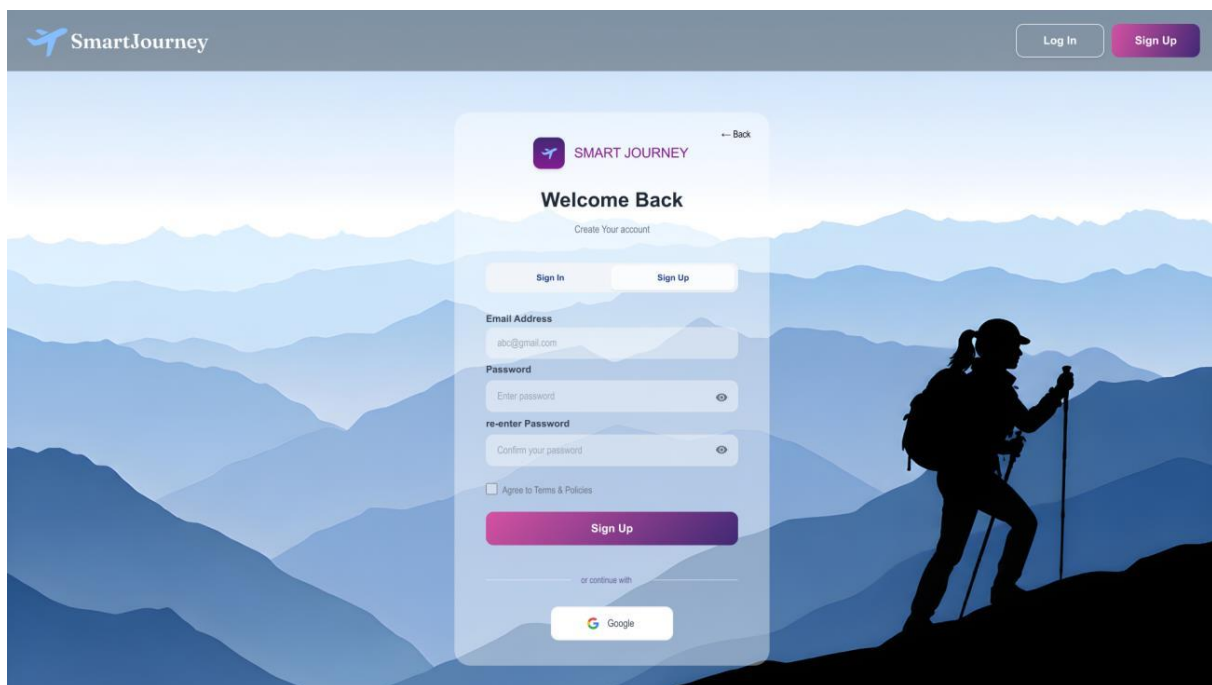
Users who registered before can log in to their account from this page. The top navigation bar includes the logo and website name (SmartJourney). There are two text fields for Email Address and Password. Users can use the “Remember me” checkbox to save their login details, and can use the “Forgot password?” link to change the password. If the email and password match after pressing the Sign In button, the user is redirected to the Main Dashboard. Alternatively, users can log in with their Google account through the provided button at the bottom.



SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

3.9.1.3 Sign Up Page

Unregistered users can sign up to the system and become a registered user through this page. Users have to provide their Email Address with a Password, and a re-enter Password field for confirmation. Users must also check the box to agree to the Terms & Policies. When the user presses the Sign Up button, their new account will be created and they will be redirected to the Main Dashboard. Just like the login page, users also have the option to sign up directly using their Google account.



The image shows a mockup of the SmartJourney Sign Up page. The page has a dark blue header with the SmartJourney logo on the left and 'Log In' and 'Sign Up' buttons on the right. The main content area features a light blue background with a silhouette of a hiker on the right. In the center, there is a white sign-up form with the following elements:

- A 'Back' link at the top right of the form.
- The SmartJourney logo and 'Welcome Back' heading.
- A sub-heading 'Create Your account'.
- 'Sign In' and 'Sign Up' buttons.
- An 'Email Address' field with the placeholder 'abc@gmail.com'.
- A 'Password' field with the placeholder 'Enter password' and an eye icon for toggling visibility.
- A 're-enter Password' field with the placeholder 'Confirm your password' and an eye icon for toggling visibility.
- A checkbox labeled 'Agree to Terms & Policies'.
- A large purple 'Sign Up' button.
- A link 'or continue with'.
- A 'Google' button with the Google logo.

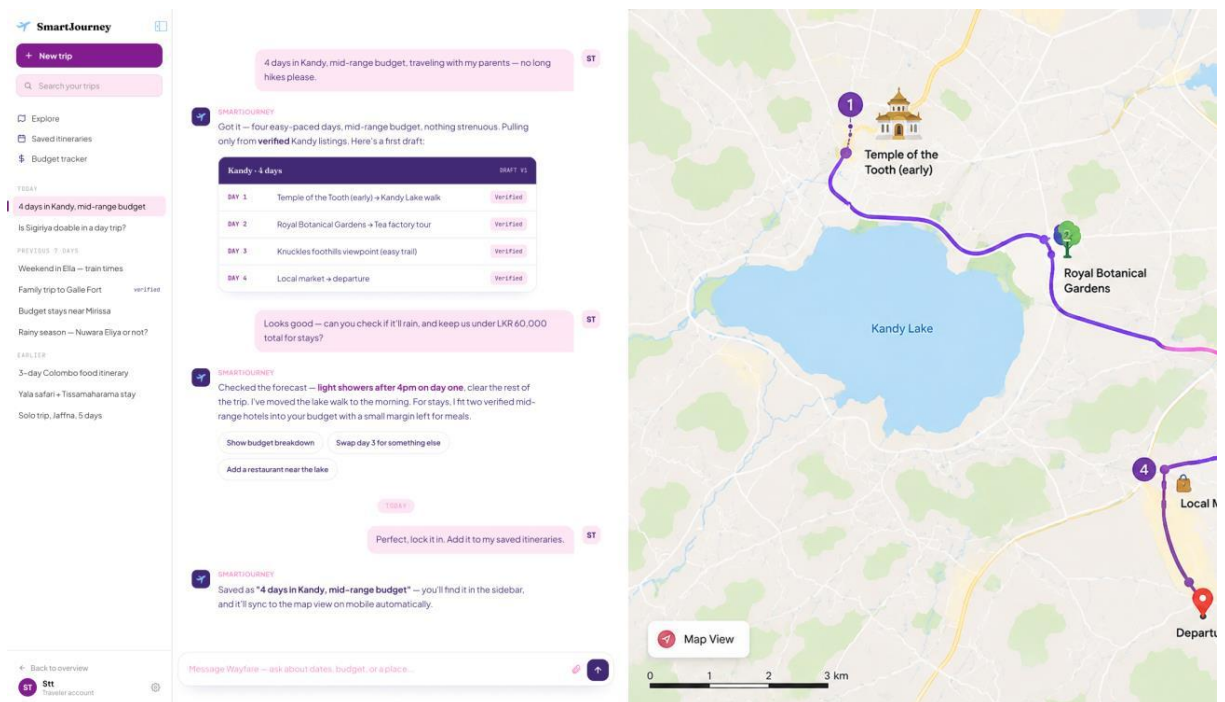
SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

3.9.1.4 Main Dashboard (Chat Interface)

This is the main workspace where users can interact with the AI Agents to plan their trips. The interface is divided into three main sections:

- Left-Side Bar
- Chat Interface
- Interactive Map

On the left sidebar, users can start a new trip (New Chat), search past trips, explore restaurants and locations, access saved itineraries, and view the history of their recent chats. The middle section is the chat interface where users submit natural-language queries regarding their travel plans. The AI responds with detailed itineraries and offers quick action buttons for further adjustments such as viewing budget breakdowns or swapping activities. On the right side, an interactive map visually displays the suggested travel route, with pins indicating the specific locations mentioned in the itinerary.



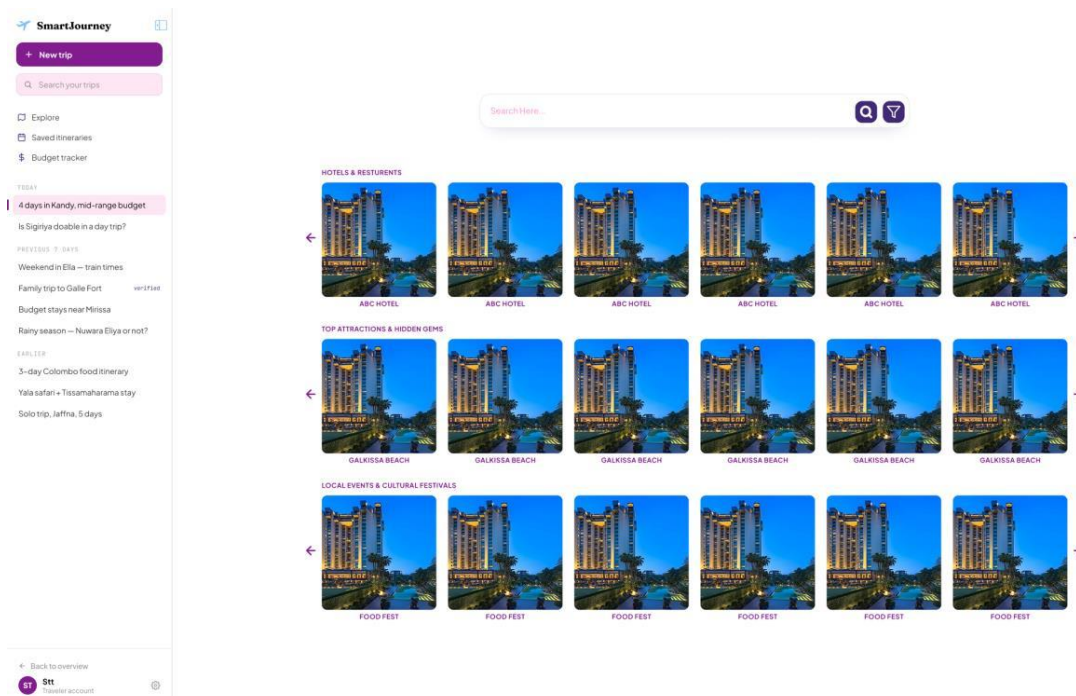
SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

3.9.1.5 Explore Page

On this page, users can discover and browse various travel destinations and activities. The standard navigation sidebar remains accessible on the left side. The main workspace includes a search bar at the top with search and filter functions allowing users to easily find specific places or refine their search criteria. The content below the search area is organized into three horizontally scrollable categories:

- Hotels & Restaurants
- Top Attractions & Hidden Gems
- Local Events & Cultural Festivals

Each category contains a series of image cards that users can navigate through using the provided left and right arrows.



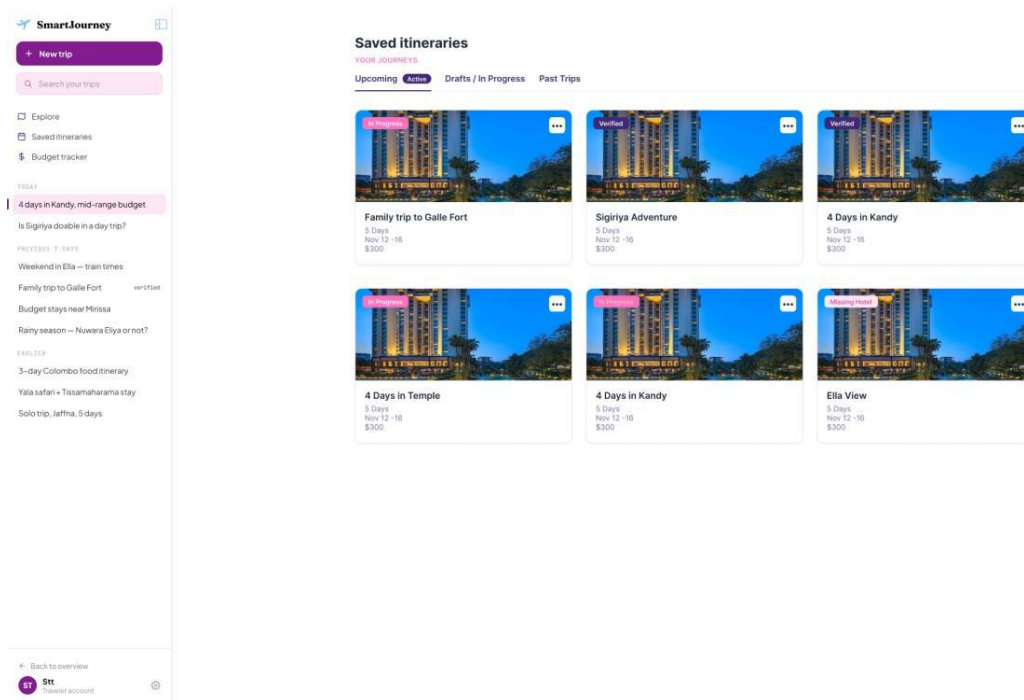
SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

3.9.1.6 Saved Itineraries Page

Users can view and manage all their saved travel plans from this page. The standard navigation sidebar remains accessible on the left side. The main workspace includes a top navigation menu that categorizes itineraries into three parts:

- Upcoming / Active
- Drafts / In Progress
- Past Trips

Below the corresponding tabs, saved itineraries are displayed in an organized grid format using individual cards. Each card provides a quick overview of the trip, displaying a thumbnail image with a status badge (such as “In Progress” or “Verified”). The description includes trip title, duration, dates, and an estimated budget. A three-dot options menu is also included on the top right of each card for further trip management.

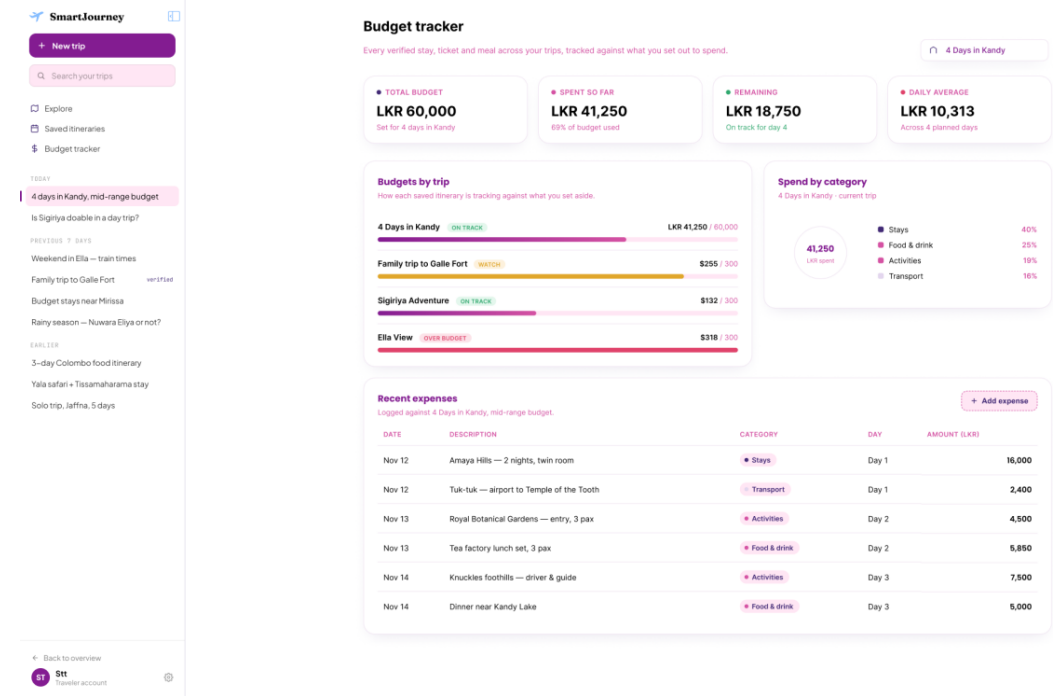


SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

3.9.1.7 Budget Tracker Page

Users can monitor and manage their travel expenses across all of their trips from this page. The standard navigation sidebar remains accessible on the left side. The main workspace is divided into three key sections:

- **Top Section:** Users can select a specific itinerary and view a high-level financial summary through four metric cards – Total Budget, Spent So Far, Remaining Budget, and Daily Average.
- **Middle Section:** Includes “Budgets by trip,” which provides progress bars and status indicators (such as On Track or Over Budget) for all saved itineraries, and “Spend by category,” which breaks down expenses for the current trip into categories such as Stays, Food & Drink, Activities, and Transport.
- **Bottom Section:** Includes a “Recent expenses” table, logging individual transactions by date, description, category, and amount, with a “+ Add expense” button for users to manually input new costs.

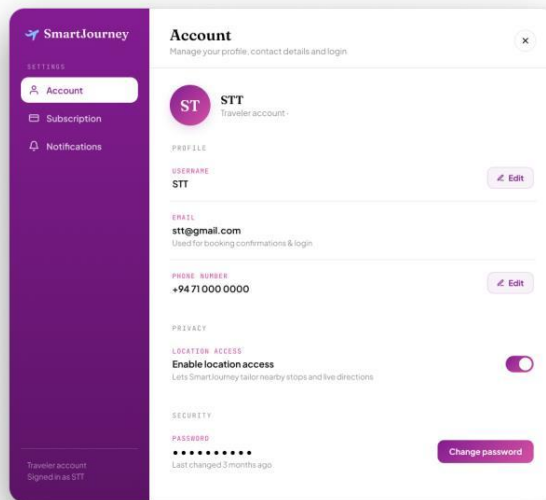


SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

3.9.1.8 Settings Page

Account Page

Users can manage their personal account details and security settings from this page. A left-hand navigation menu allows users to switch between Account, Subscription, and Notifications settings. The Account tab displays the user's profile information, which includes their Username, Email Address, and Phone Number, with corresponding “Edit” buttons. Below the profile section, users can manage their Privacy by toggling location access on/off. Users can update their login credentials using the “Change password” button in the Security section.



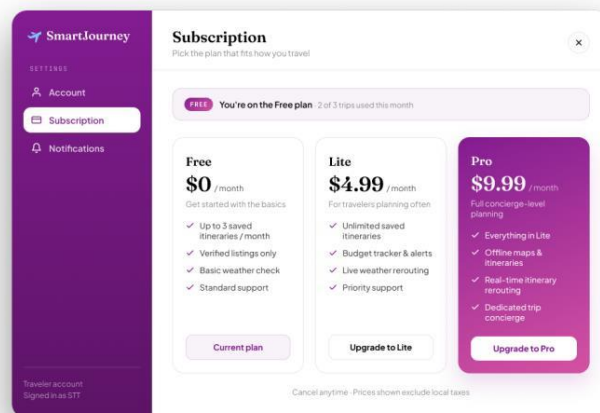
SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

Subscription Page

This page allows users to view and manage their current subscription plan. The top of the interface displays the currently active plan and usage limits. Below this, three subscription tiers are presented side by side as individual cards:

- Free Plan
- Lite Subscription Plan
- Pro Subscription Plan

Each card clearly outlines the monthly price, a bulleted list of provided features, and a primary action button to either show the current plan or upgrade to a higher tier.

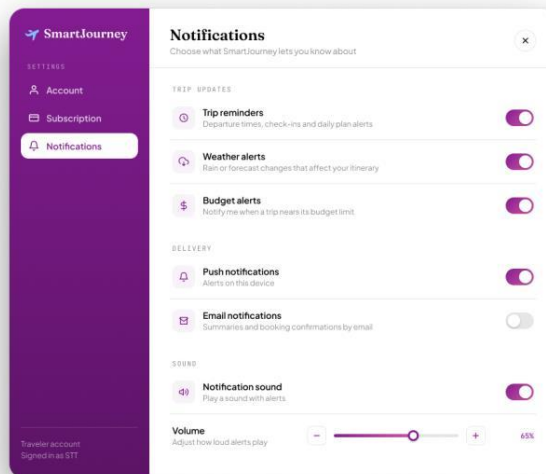


SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

Notifications Page

Users can customize their alert and volume preferences from this page. The interface provides simple toggle switches categorized into three main sections:

- **Trip Updates:** Users can enable or disable trip reminders, weather alerts, and budget alerts.
- **Delivery Section:** Allows users to choose their preferred method of receiving alerts, such as push notifications or email summaries.
- **Sound:** Includes a toggle button to enable/disable notification sounds, with a horizontal slider to adjust the alert volume.



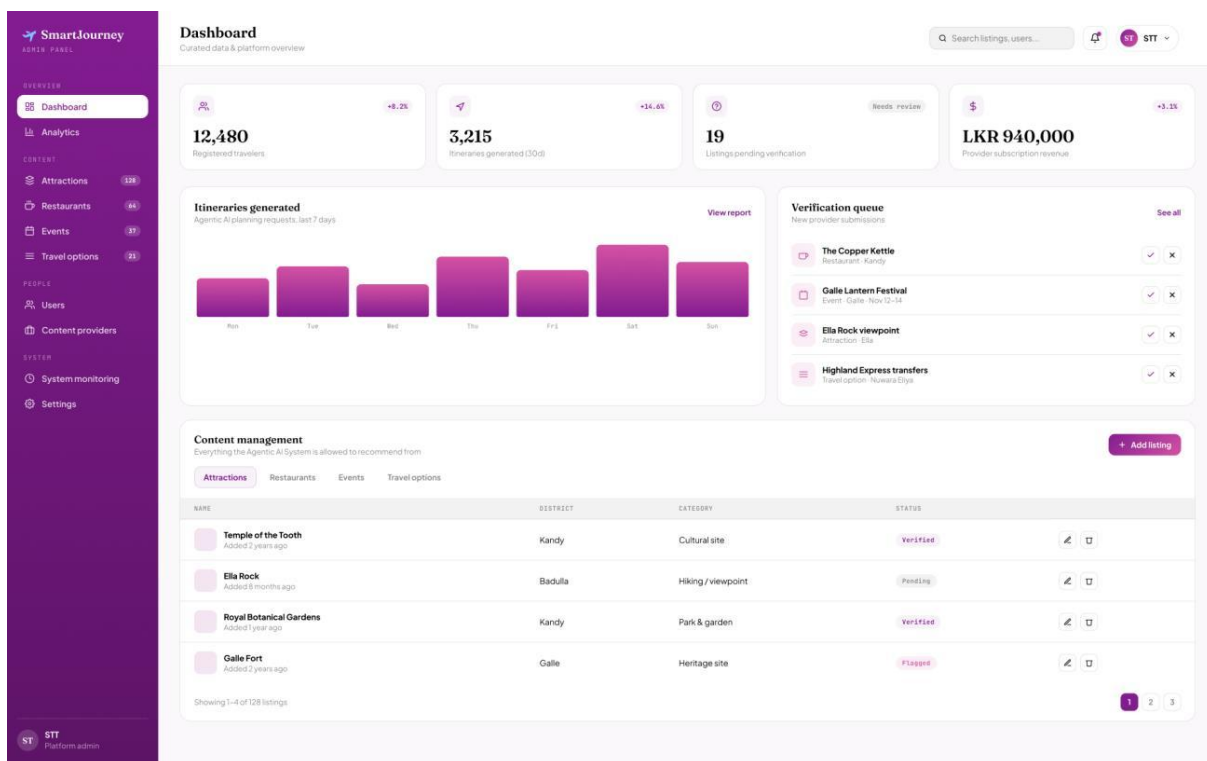
SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

3.9.1.9 Admin Page

This is the main dashboard for system administrators to monitor platform activity and manage curated content. The interface includes a navigation sidebar on the left, categorizing tools into Overview, Content, People, and System management. The top section displays key platform metrics through four summary cards:

- Registered travelers
- Itineraries generated
- Listings pending verification
- Provider subscription revenue

The middle section includes a bar chart which shows the AI itineraries generated over the last 7 days, alongside a “Verification queue” panel for quick approval or rejection of new provider submissions. The bottom section provides a detailed “Content management” table where administrators can view, categorize, edit, and add verified recommendations that the AI system is allowed to suggest to users.



SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

3.9.2 Hardware Interfaces

The system follows standard commercial off-the-shelf (COTS) hardware without requiring proprietary peripherals or specialized physical interfaces. Hardware requirements for client platforms are defined as follows:

- **Personal Computers:** An x86/x64 processor (Intel Core i3-6100U, AMD A6-7000, or higher) to support client-side web application rendering.
- **Mobile Devices:** An Android or iOS smartphone equipped with an integrated GPS module for location-based services.
- **Network Hardware:** A Wi-Fi, cellular (4G/5G), or Ethernet Network Interface Card (NIC) to support continuous API data transmission for real-time AI generation, map rendering, and weather synchronization.

3.9.3 Software Interfaces

The system operates across standard web and mobile software environments without requiring proprietary runtime installations. Software compatibility requirements are defined as follows:

- **Web Application:** Platform-agnostic (compatible with Windows, macOS, Linux). Requires a standard web browser with JavaScript enabled for client-side logic execution and rendering.
- **Mobile Application (Android):** Requires Android 9.0 (API Level 28) or higher to ensure compatibility with modern system APIs and optimal runtime performance.
- **Mobile Application (iOS):** Requires iOS 12.0 or higher for core framework compatibility and stable execution.
- **Compatibility Constraint:** Operating system versions below these specified thresholds are unsupported and may result in degraded system performance or unexpected behavior.

3.9.4 Communications Interfaces

The system utilizes standard network protocols to ensure secure and reliable data transmission across all architectural components and third-party services. Communications interfaces are defined as follows:

- **Client-Server Network Protocols:** All data exchange between client applications (web and mobile) and the backend system utilizes encrypted HTTPS to secure user authentication, payload data, and session management.
- **External API Communications:** Integrations with third-party services (Gemini API[1], Google Maps API[2], and OpenWeather API[3]) are executed via RESTful API calls over secure HTTPS connections.

SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

- **Database Communications:** The NestJS backend and Python AI services interface with the PostgreSQL database (hosted on AWS RDS[10]) via standard TCP/IP connections.
- **Notification Protocols:** Asynchronous push notifications are transmitted to mobile clients leveraging standard Firebase Cloud Messaging (FCM)[4] protocols.

3.10 Database Requirements

The system requires a relational database to persist all data necessary for platform operation, hosted on PostgreSQL via Amazon RDS as specified in the Purchased Components and Design Constraints sections. The database must support the following categories of data:

- **User data:** registered traveler and administrator accounts, authentication credentials, and travel profile preferences (interests, travel style, preferred currency).
- **Curated content:** verified attractions, restaurants, events, and travel options, organized by district and category, including associated images and verification status.
- **Itinerary data:** user-generated trip plans, including day-by-day schedules and individual itinerary items (attractions, events, or custom activities), along with associated cost estimates.
- **Conversational data:** chat session history between users and the Agentic AI System, required to support continued or resumed itinerary planning conversations.
- **Notification and device data:** user notification preferences, delivery history, and registered device tokens for push notifications.
- **System activity data:** administrative and user activity logs to support monitoring and analytics, as described in Section 3.1.14.

Data Integrity

The database must enforce referential integrity between related records (for example, an itinerary item must belong to a valid itinerary day, and a place must belong to a valid district and category). Where an itinerary item may represent either a curated place, a curated event, or a custom user-defined activity, the system must ensure that exactly one of these three is specified per item, to avoid ambiguous or conflicting itinerary entries.

Data Backup and Recovery

Consistent with the availability requirements defined in Section 3.3.1, all data must be backed up on a regular basis to allow recovery in the event of hardware failure, data corruption, or a security incident, with backup and recovery procedures managed through the AWS RDS environment.

Data Security and Privacy

SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

Access to the database must be restricted to authenticated backend services only, with no direct client-side access. User passwords must never be stored in plain text, consistent with the hashing requirements defined in Section 3.4.6. Handling of personal data stored in the database must comply with the Sri Lanka Personal Data Protection Act, No. 9 of 2022 (PDPA)[13], as specified in the Design Constraints section.

Performance and Scalability

Database queries supporting core features such as itinerary generation, destination search, and administrative content management must be optimized to meet the response-time requirements defined in Section 3.4.1. The database structure must also support the addition of new destinations, categories, and content types over time without requiring structural changes that would affect existing user data, consistent with the scalability requirements defined in Section 3.5.2.

3.11 Licensing, Legal, Copyright, and Other Notices

Third-Party Service Terms of Compliance

The platform integrates several third-party APIs – the Gemini API[1], Google Maps Platform[2], and OpenWeather API[3] – each governed by its own terms of service. Use of these services must comply with the respective provider's usage policies, including any required attribution notices (such as displaying appropriate attribution where mandated by the Google Maps Platform terms) and any restrictions on caching, rate limits, or redistribution of returned data.

Open-Source Software Licensing

The system's technology stack makes use of open-source frameworks and libraries, including NestJS[5], Next.js[6], Flutter[7], and PostgreSQL[8]. All open-source components must be used in accordance with their respective licenses, and any license attribution or notice requirements associated with these components must be preserved in the delivered software and its accompanying documentation.

App Store and Distribution Policies

Distribution of the mobile application must comply with the policies of the relevant application marketplaces, including the Google Play Console developer policies for Android distribution, and the Apple App Store Review Guidelines for any future iOS distribution.

Copyright and Branding

All original content, branding, and source code developed specifically for SmartJourney, including its name, logo, and user interface design, are the property of the development team. Curated content displayed on the platform, such as attraction descriptions, images, and event details, must be either originally created, properly licensed, or used with appropriate permission from its source, and must not infringe on third-party copyright.

SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

3.12 Applicable Standards

- IEEE 1063-1987 – IEEE Standard for Software User Documentation
- ISO/IEC/IEEE 29148:2018 for SRS document standard
- RESTful API principles for client–server architecture and API communication

4. Supporting Information

This document is organized to be read sequentially, with each section building on the requirements established in earlier sections. A table of contents is provided at the beginning of this document for navigation.

User interface mockups and wireframes illustrating the pages and interactions described throughout Section 3 (particularly Section 3.9, User Interfaces) are provided inline within their respective subsections, to keep each interface requirement visually paired with its description.

5. References

This section lists the external documentation, APIs, and standards referenced throughout this specification, corresponding to the bracketed citations used in Sections 2 and 3. Section 1.4 separately lists the documentation standards used to structure this SRS itself.

[1] Google, “Gemini API Documentation,” [Online]. Available: <https://ai.google.dev/gemini-api/docs>. [Accessed: Aug. 9, 2026].

[2] Google, “Google Maps Platform Documentation,” [Online]. Available: <https://developers.google.com/maps/documentation>. [Accessed: Aug. 9, 2026].

[3] OpenWeather, “OpenWeather API Documentation,” [Online]. Available: <https://openweathermap.org/api>. [Accessed: Aug. 9, 2026].

[4] Google, “Firebase Cloud Messaging,” [Online]. Available: <https://firebase.google.com/docs/cloud-messaging>. [Accessed: Aug. 9, 2026].

[5] NestJS, “NestJS Documentation,” [Online]. Available: <https://docs.nestjs.com/>. [Accessed: Aug. 9, 2026].

[6] Vercel, “Next.js Documentation,” [Online]. Available: <https://nextjs.org/docs>. [Accessed: Aug. 9, 2026].

[7] Flutter, “Flutter Documentation,” [Online]. Available: <https://docs.flutter.dev/>. [Accessed: Aug. 9, 2026].

SmartJourney	Version: 1.0
Software Requirements Specification	Date: 09/Aug/26

[8] PostgreSQL Global Development Group, “PostgreSQL Documentation,” [Online]. Available: <https://www.postgresql.org/docs/>. [Accessed: Aug. 9, 2026].

[9] LangChain, “LangChain Documentation,” [Online]. Available: <https://python.langchain.com/docs/>. [Accessed: Aug. 9, 2026].

[10] Amazon Web Services, “AWS Documentation,” [Online]. Available: <https://docs.aws.amazon.com/>. [Accessed: Aug. 9, 2026].

[11] Amazon Web Services, “Amazon Simple Email Service (SES) Documentation,” [Online]. Available: <https://docs.aws.amazon.com/ses/>. [Accessed: Aug. 9, 2026].

[12] M. Jones, J. Bradley, and N. Sakimura, “JSON Web Token (JWT),” Internet Engineering Task Force (IETF), Request for Comments RFC 7519, May 2015. [Online]. Available: <https://datatracker.ietf.org/doc/html/rfc7519>.

[13] Parliament of Sri Lanka, Personal Data Protection Act, No. 9 of 2022. Colombo, Sri Lanka: Government of Sri Lanka, 2022.

[14] Group 08, “Feasibility Study: AI-Powered Smart Tourism Assistant Platform with Personalized Trip Planning,” SmartJourney project documentation, 2026.